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Master's Programme in Advanced Energy Solutions

The impact of battery energy storage on day-ahead electricity prices

A price-based analysis of arbitrage and charging-cost regulation in the German power market

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Aalto University

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Title of thesis The impact of battery energy storage on day-ahead electricity prices

Programme Advanced Energy Solutions

Major Energy systems and markets

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Date 01.02.2026 **Number of pages** 45 **Language** English

Abstract

The increasing penetration of variable renewable energy sources has intensified short-term price volatility and intertemporal price spreads in European electricity markets, increasing the value of flexibility resources capable of shifting electricity across time. Battery energy storage systems (BESS) are emerging as a key source of short-duration flexibility, yet their influence on day-ahead electricity prices remains insufficiently quantified, particularly under alternative regulatory charging-cost structures. This thesis develops a price-based analytical framework to assess how large-scale BESS operating through day-ahead arbitrage modify electricity price formation in the German market. The methodology combines a profit-maximising, price-taking battery arbitrage optimisation model with an iterative price-adjustment mechanisms that updates hourly prices conditional on battery charging and discharging behaviour. Using historical day-ahead prices for four representative weeks from 2023-2024, the analysis focuses on price dynamics rather than storage profitability. Results show that BESS arbitrage reduces price extremes, compresses intertemporal price spreads, and lowers short-term volatility, with the magnitude of these effects strongly dependent on charging-cost structures. The findings highlight the role of charging-cost-regulation in shaping storage-induced price dynamics in high-renewable power systems.

Keywords German electricity market, Spot, Energy arbitrage, Flexibility, Price impact, charging-cost regulation

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Symbols and abbreviations

aFRR - Automatic Frequency Restoration Reserve
BESS - Battery energy storage systems
CAPEX – Capital Expenditure
DA - Day Ahead
DE – Germany
EUA – European Union Allowance
FCR - Frequency Containment Reserve
SATI – Strategic Assessment tool for Investment
SoC - State of charge
VAT - Value-added Tax
VRES - Variable Renewable Energy Sources

1 Introduction

As the share of variable renewable energy sources (VRES) continues to increase in pursue of a decarbonised European power system, electricity markets are experiencing growing short-term price volatility and more pronounced intertemporal price spreads. The expansion of wind and solar generation amplifies fluctuations in residual load, placing increasing value on flexibility resources capable of shifting electricity across time (Brijs et al., 2019). In this context, wholesale electricity prices, particularly in the day-ahead (DA) market, have become a key signal for balancing short-term supply and demand under high renewable penetration.

Battery energy storage systems (BESS) have emerged as a central source of short-duration flexibility due to their rapid response capabilities, modularity, and declining costs. Utility-scale BESS are increasingly deployed to support system operation and to participate in electricity markets, complementing other flexibility options such as dispatchable generation and demand response (Mercier et al., 2023). From an operational perspective, battery assets can engage in multiple market segments, including but not limited to balancing services, intraday trading, and energy arbitrage in the day-ahead market, with the chosen revenue mix strongly influenced by both market conditions and regulatory frameworks.

A substantial body of literature has investigated the value of electricity storage from both the system and investor perspectives. Early studies often relied on perfect foresight and price-taking assumptions, evaluating storage arbitrage revenues using exogenous price time series and implicitly assuming that storage deployment is too small to influence market prices (Kazempour et al., 2009; Zafirakis et al., 2016). More recent studies have relaxed these assumptions by incorporating the price effect of storage, showing that storage charging and discharging can smooth price profiles and reduce intertemporal price spreads as storage installed capacity increases. These studies highlight a cannibalization effect, whereby arbitrage revenues decline as storage deployment expands (Brijs et al., 2019; Mercier et al., 2023).

Despite these advances, the literature provides limited insight into how battery storage operations modify day-ahead electricity prices themselves, as distinct from how profitability evolves with deployment. While several studies document declining arbitrage revenues under endogenous pricing, fewer explicitly analyse the resulting changes in the day-ahead price curve independently of investment cost or revenue assumptions.

Although current installed utility-scale BESS capacity in Germany is not yet large enough to exert a measurable influence on wholesale price formation,

the market exhibits high renewable penetration, frequent price volatility, and growing interest in large-scale battery deployment. These characteristics make Germany a suitable setting for analysing how battery arbitrage could influence day-ahead prices as deployment scales.

In addition to market fundamentals, regulatory design plays a critical role in shaping battery operation and its price effects. In Germany, electricity storage benefits from specific regulatory exemptions affecting the treatment of electricity withdrawn for charging, including exemptions from network charges, regulated levies and surcharges under defined conditions. These regulatory provisions materially alter the effective charging costs faced by BESS and, in turn, influence optimal arbitrage behaviour. As a result, different regulatory charging-cost structures can lead to distinct battery dispatch patterns and heterogeneous impacts on day-ahead prices.

This thesis develops a price-based analytical framework to assess how battery arbitrage operation modifies day-ahead electricity prices under alternative regulatory charging-cost structures. Using VaasaETT's proprietary battery optimization tool, SATI (Strategic Assessment Tool for Investment), the methodology constructs modified day-ahead price curves by iteratively adjusting peak and trough prices through the minimum necessary changes. The analysis focuses on isolating the **price effects induced by battery arbitrage**, conditional on three distinct regulatory structures derived from existing exemptions on battery charging costs.

The empirical analysis focuses on the German electricity market, reflecting its advanced renewable integration, evolving regulatory framework for storage, and relevance for future large-scale BESS deployment. By quantifying how regulatory charging-cost structures influence battery arbitrage behaviour and its impact on day-ahead price formation, this thesis contributes to a clearer understanding of the interaction between storage regulation, market operation, and wholesale price dynamics. The findings are relevant for policymakers, regulators, system operators, and market participants engaged in the design of electricity markets and flexibility frameworks in power systems with high share of variable renewable energy.

2 Literature review

This section reviews the academic literature relevant to analysing how battery energy storage systems interact with electricity prices in the day-ahead market. The reviewed studies provide the theoretical and empirical basis for understanding storage arbitrage, price feedback effects, and the resulting impact of storage deployment on wholesale price formation. The section identifies the limitations of existing approaches and motivates the price-based framework developed in this thesis to assess the impact of battery storage on day-ahead electricity prices.

2.1 Price formation in high-renewable electricity markets

This section reviews the key mechanisms through which high renewable penetration reshapes wholesale electricity prices and creates the conditions under which short-duration storage can extract value through arbitrage.

2.1.1 Merit order effect

As shown in numerous empirical studies, the expansion of wind and solar generation has altered electricity altered electricity price formation by displacing higher marginal-cost thermal units in the merit order. Liebensteiner et al. (2025) demonstrate that increasing renewable penetration in Germany and other European markets has reduced average wholesale prices while simultaneously increasing price variability.

Germany provides a mature example of this transition. According to official statistics (SMARD, 2025), renewable energy sources accounted for approximately 59% of total net electricity generation in 2024, up from less than 30% in 2010, with wind and solar as the dominant contributors (AGEE/BMWK, 2025). As a result, hourly price formation in the German day-ahead market is increasingly driven by short-term weather-dependent supply conditions rather than fuel-based marginal costs.

Figure 1 illustrates this mechanism by showing the relationship between hourly renewable generation and day-ahead prices in Germany, highlighting the role of variable supply in shaping short-term price dynamics.

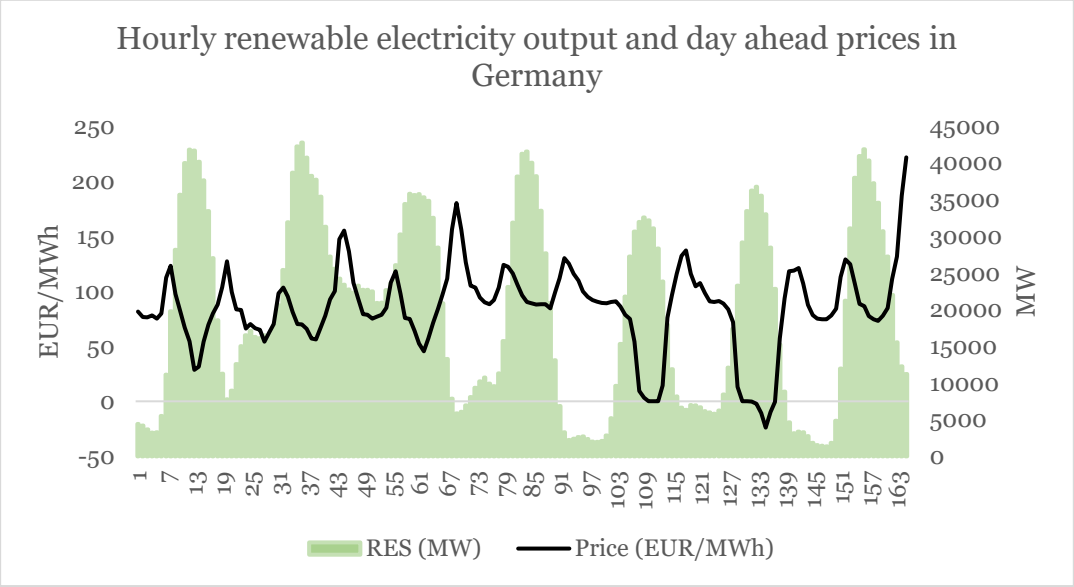


Figure 1. Relationship between hourly renewable generation and day-ahead electricity prices in Germany (representative week 21, 2024) (Sources: Energy Market Price, 2025; ENTSO-E, 2025)

2.1.2 Volatility and spread reshaping

While higher RES penetration reduces average wholesale prices, the literature consistently finds that it increases inter-hour price volatility. Liebensteiner et al. (2025) show that wind and solar output significantly increase price variance rather than uniformly depressing prices. Prokhorov and Dreisbach (2022) further demonstrate that high renewable output rises the probability of very low prices, while residual load, gas price and CO2 allowance price dominate price formation during scarcity hours.

This combination of intermittent supply and limited short-term flexibility leads to wider inter-hour price spreads, which are necessary conditions for profitable storage arbitrage.

2.1.3 Negative prices and curtailments as structural outcomes

A direct manifestation of these dynamics is the growing incidence of negative electricity prices. In Germany, negative prices have been allowed in the day-ahead market since 2007 as a market-based signal for flexibility scarcity. Their frequency has increased sharply with renewable deployment.

In 2024, the number of negative priced hours reached 457, up from 301 hours in 2023 and fewer than 100 hours in the early 2010's (SMARD). The minimum day-ahead price in 2024 reached -135.45 EUR/MWh, while peak price was 936.28 EUR/MWh, illustrating the dispersion of prices.

Curtailment of renewable generation is closely linked to these outcomes. When grid or system constraints prevent full absorption of renewable output, generation must be curtailed despite low marginal costs, representing an economic inefficiency. Both negative prices and curtailment are therefore increasingly interpreted in the literature as structural indicators of insufficient system flexibility, rather than transient market anomalies (Liebensteiner et al., 2025).

2.1.4 Flexibility scarcity creates arbitrage opportunities

The coexistence of frequent low or negative prices during surplus periods and high prices during scarcity hours generates persistent temporal price spreads. In the absence of sufficient flexibility, these spreads remain and may intensify as renewable penetration increases. From an economic perspective, such price structures create incentives for technologies capable of shifting electricity across time.

2.2 Battery storage and arbitrage in day-ahead markets

The price patterns described above provide the economic basis for battery storage operation. This section reviews how BESS exploits intertemporal price differences and explains why the day-ahead market is the appropriate focus for analysing price effects.

2.2.1 Role of BESS as intertemporal arbitrage

Battery energy storage systems provide flexibility by shifting electricity from low-price periods to high-price periods. Metz and Saraiva (2018) formalise this mechanism, showing that storage arbitrage depends on price spreads rather than absolute price levels.

It is important to distinguish between deployment scales. While residential batteries dominate in unit numbers in Germany, utility-scale BESS, with are the focus of this study, have historically derived most revenues from balancing markets in Germany, However, recent market developments indicate a growing shift towards intraday and day-ahead arbitrage as battery capacities increase and price volatility intensifies (Fraunhofer ISE, n.d).

2.2.2 Day-ahead as the focus market

The day-ahead market is the primary point of wholesale price formation in Germany. It internalises system wide expectations regarding renewable

generation, demand, and cross-border exchanges, making it particularly suitable for analysing structural price effects. In 2024, German electricity consumption was approximately 510 TWh, of which around 430-450 TWh (85-90%) were cleared in the day-ahead market.

Renewable-driven price phenomena, including negative prices and scarcity peaks, are most clearly observable in day-ahead prices. From a modelling perspective, day-ahead prices are less affected by forecast errors than intra-day prices, allowing storage operation to be analysed as an endogenous price-modifying activity.

2.2.3 Key determinants of arbitrage value

The literature identifies three primary determinants of storage arbitrage value. Taken together, these factors define the economic envelop withing which battery arbitrage can affect prices.

First, price spreads. Arbitrage revenues depend primarily on the magnitude and frequency of price spreads. Empirical studies show that higher renewable penetration increases the occurrence of low-price periods, thereby expanding potential arbitrage opportunities in the short term (Liebensteiner et al., 2025; Prokhorov and Dreisbach, 2022).
Second, efficiency. Roundtrip efficiency determines how much of the charged energy can be shifted across time. Lithium-ion batteries, which dominate current BESS deployments, typically exhibit round-trip efficiencies between 85-95%, as reported in modelling studies.

Third, duration. Storage duration determines the amount of energy that can be shifted but exhibits diminishing marginal returns. Studies show that increasing storage duration beyond a few hours yields progressively smaller incremental arbitrage revenues, particularly in systems where price volatility is driven by short-term renewable fluctuations rather than prolonged scarcity events.

This thesis focuses exclusively on day-ahead energy arbitrage. Revenues from ancillary services (e.g. Frequency Containment Reserve (FCR), automatic Frequency Restoration Reserve, aFRR) or intraday trading are excluded. Germany does not operate a centralized capacity mechanism; therefore, capacity market revenues are not considered relevant for BESS in the German context. These exclusions allow for a clearer isolation of the price-storage interaction in the day-ahead market.

2.3 Price impact and arbitrage saturation of battery storage

Previous sections established why intertemporal price spreads emerge in high-renewable electricity systems and how battery energy storage system exploit them.

2.3.1 Price taking vs price-making storage

Early studies on storage arbitrage typically assume batteries to be price takers, implicitly assuming that installed storage is too small to affect market clearing prices. For example, early optimization-based studies evaluate revenues by applying exogenous price time series to optimal charging and discharging strategies.

While this assumption may be reasonable for small-scale or early-stage deployment, it becomes invalid when utility-scale BESS installed capacity increases. Once aggregated charging and discharging volumes become non-negligible relative to system demand, storage operation directly affects the supply-demand balance in individual hours.

This effect has been explicitly demonstrated in studies that endogenies electricity prices. Metz and Saraiva (2018) show that when battery operation is modelled withing the day-ahead market-clearing process, storage charging rises price in low-price hours, while discharging reduces prices during high-price hours. Similarly, Veenstra and Mulder (2025) demonstrate that ignoring this feedback leas to systematic overestimation of arbitrage revenues. These findings confirm that large-scale BESS should be treated as price-making assets rather than price takers when analysing their impact on day-ahead prices.

2.3.2 Spread compression, peak shaving, and cannibalization effects

The price impacts of battery storage arise from two symmetric mechanisms. During low-price periods, charging increases demand and raises prices; during high-price periods, discharging increases supply and reduces prices. AS shown in endogenous-price models, these mechanisms lead to spread compression, commonly described as valley filling and peak shaving.

Studies show that increasing storage capacity in the day-ahead market systematically reduced both extreme low and extreme high prices, leading to a narrower price distribution. Metz and Saraiva (2018) reach similar conclusions, demonstrating that storage smooths price profiles by reducing intra-hour price differentials.

This price-smoothing effect gives rise to the cannibalization of arbitrage revenues. As more storage capacity is installed, the price spreads that enable arbitrage are progressively reduced by storage operation itself. Lamp & Samano (2022) show that arbitrage revenues decline non-linearly with installed storage capacity when prices are endogenous, while Rangarajan et al. (2023) confirm that marginal revenues decrease as storage penetration increases.

Importantly, these effects are structural: once storage deployment reaches system-relevant levels, additional capacity yields diminishing price effects.

2.3.3 Limitations of existing approaches

While the literature clearly establishes that battery storage modifies day-ahead prices and reduces arbitrage revenues as deployment increases, most studies focus on profitability outcomes rather than isolating the price effects of storage operation. In particular, existing analyses typically report declining revenues for predefined storage capacities without explicitly characterising how the underlying price structure changes as a result of storage deployment.

Moreover, although several studies examine whether arbitrage revenues diminish below cost-recovery level, they do not provide a price-based criterion for assessing how storage operation modifies the day-ahead price profile itself. As a result, the literature offers limited guidance on quantifying the extent to which battery deployment alters price formation independently of investment cost assumptions.

This gap motivates the methodological approach adopted in this thesis, which focuses on identifying the price effects induced by battery arbitrage in the day-ahead market.

2.4 Electricity charging costs for BESS in Germany

This section describes the regulatory framework governing the costs incurred when BESS withdraw electricity from the grid in Germany. The objective is to identify which costs components apply to battery charging, which exemptions and reductions exist under current legislation, and which charging-costs treatments are relevant for the modelling framework applied in this thesis. The section is descriptive and regulatory in nature and provides the foundation for the charging-sots cases defined in Section 3.3.

2.4.1 Regulatory classification of electricity storage

Under German energy law, electricity storage facilities are legally classified as both final consumers and generators. When charging, a battery withdraws electricity from the grid and is treated as a consumer. When discharging, it feeds electricity back into the grid and is treated as a generator. This dual classification is explicitly recognised in the Energy Industry Act (EnWG) and reflects the technical function of storage as a controllable temporal shift between electricity generation and consumption.
This classification implies that electricity withdrawn for charging is, in principle, subject to network usage charges, regulated levies and taxes, while electricity injected during discharging is not subject to such charges. At the same time, the regulatory framework provides **specific exemptions** to avoid double charging of electricity that is temporarily stored and later re-fed into the grid, recognising the system-supporting role of storage.

2.4.2 Network usage charges and construction costs

Network usage charges constitute a major component of electricity withdrawals costs in Germany. Large-scale electricity storage facilities may benefit from exemptions from network usage charges pursuant to § 118(6) EnWG, subject to regulatory approval and compliance with eligibility requirements. These exemptions are intended to support storage operation that contributed to system stability and efficient grid utilisation and may include exemptions for storage-related losses.

In addition to operational network usage charges, grid-connected storage facilities may be required to pay construction cost contributions (Baukostenzuschüsse, BKZ) when connecting to the grid. Recent regulatory developments and court decisions have clarified that transmission systems operators may grant substantial reductions in BKZ, particularly in regions where battery deployment supports grid operations. While BKZ represents an important locational investment signal, it constitutes a one-off connection cost and do not affect short-term operational decisions. Accordingly, BKZ is not considered in the arbitrage modelling performed in this thesis.

2.4.3 Regulated levies and surcharges on electricity withdrawals

In addition to network usage charges, electricity withdrawals in Germany are subject to several regulated levies (Umlagen), which are applied nationwide and defined by federal legislation. During the study period (2023-2024), the relevance of these levies for electricity storage is as follows:

- **§ 19 (2) StromNEV** - Surcharge for special grid utilisation. A regulated levy collected via electricity bills to finance reductions in network usage charges granted to certain customers. Electricity withdrawn for battery storage is subject to this levy by default.

- **§ 17f EnWG** - The offshore grid levy – Applies to electricity withdrawals but it subject to storage-specific limitations.
- **§ 27b KWKG** – The Combined Heat and Power levy - Applies to electricity withdrawals but it subject to storage-specific limitations.
- **§ 18 AbLaV** – A surcharge abolished and no longer charged from 2023 onwards
- **EEG** – A levy abolished in 2022, therefore not applicable during the study period.

For electricity storage systems, the applicability of several levies is limited under § 21 of the Energy Financing Act (EnFG). This provision introduces and offsetting mechanisms whereby electricity volumes withdrawn for storage and later re-fed into the grid are largely exempt from levies to prevent double charging. The application of this mechanism requires appropriate metering and documentation of electricity flows.

2.4.4 Taxes and concession fees

Electricity withdrawn for battery charging may also be subject to electricity tax and value-added tax (VAT). Under § 5(4) of the Electricity Duty Act (StromStG), electricity storage facilities benefit from exemptions from electricity tax for electricity that is later fed back into the grid. VAT applies generally to electricity consumption but does not affect relative arbitrage incentives and is therefore not treated as a differentiating factor in the modelling framework.

The concession fee, paid to municipalities for the use of public land by electricity networks, remains payable for electricity withdrawals and represent and unavoidable component of charging costs, even where other charges are exempted

2.4.5 Summary of charging-cost treatments relevant for modelling

The German regulatory framework applies different cost components to electricity storage when withdrawn from the grid, reflecting the dual role of storage as both consumer and generator. For the purpose of this study, the relevant distinction is which grid-related charges, regulated levies, and taxes apply to electricity used for battery charging.

Based on the regulatory provisions and exemptions mechanism applicable during the study period, a small number of billing configurations for battery charging are defined. These configurations are implemented in Section 3.3 and enable a structured comparison of battery operation under different regulatory cost conditions.

Table 1 summarises the electricity billing components applicable to battery charging in Germany during the study period and indicates which components are varied across the charging-cost configurations implemented in Section 3.3

Table 1. Summary of electricity billing components applicable to BESS charging in Germany (2023-2024)

Cost component	Applicable to battery charging by default?	Storage-specific exemption/reduction	Legal basis/source	Included in modelling cases
Wholesale electricity price (DA)	Yes	No	EPEX / Nord Pool market rules	Always
Network usage charges	Yes	Partial or full exemption possible for storage, including storage losses	EnWG §118(6)	Scenario-dependent
Construction cost contribution (BKZ)	Yes (one-off, connection-related)	Significant reductions possible, region-specific	BNetzA; BGH decision 20/1653	Not modelled (CapEx)
§19 StromNEV levy	Yes	Limited for storage under offsetting mechanism	StromNEV §19; EnFG §21	Scenario-dependent
Offshore grid levy	Yes	Exempted for electricity stored and re-fed	EnFG §21	Scenario-dependent
CHP levy (KWKG)	Yes	Exempted for storage and storage losses	KWKG; EnFG §21	Scenario-dependent
EEG levy	No (abolished)	Not applicable since 2022	BMWK / BNetzA	Not applicable
AbLaV levy	No (abolished)	Not applicable since 2023	BNetzA	Not applicable
Electricity tax	Yes	Exempted for electricity re-fed from storage	StromStG §5(4)	Scenario-dependent
Concession fee	Yes	No exemption	KonzessionsabgabenVO	Always
VAT	Yes	No storage-specific exemption	German VAT law	Neutral, not modelled

3 Research material and methods

The approach chosen for this thesis to evaluate the effect of BESS in the wholesale day-ahead price is through the analysis of historical hourly price data via a VAASAETT proprietary battery profit maximization tool named SATI.

The analysis of historical hourly prices provides insights on how arbitrage opportunities in this market have developed in the last 2 years, thereby helping to estimate how these opportunities will play out in the mid-term (2030-2040) when other market trends are considered, for instance, the future composition of the power generation mix and the rising prices of carbon emissions and natural gas.

3.1 Methodology for quantifying price changes due to flexible assets in the system

This section presents the methodology used to quantify electricity price changes arising from the operation of battery energy storage systems (BESS) in the German electricity market. As discussed in Section 2, BESS in Germany can participate in multiple market segments, including the day-ahead market, intraday market, and balancing and ancillary service markets. To maintain analytical clarity and isolate the direct interaction between storage operation and price formation, this thesis focuses exclusively on day-ahead energy arbitrage, abstracting from other revenue streams and market layers.

The methodological framework builds on the economic principle that batteries exploit intertemporal price differences by charging during low-price periods and discharging during high-price periods. When deployed at sufficient scale, however, this behaviour feeds back into market prices by increasing demand during charging hours and increasing supply during discharging hours, thereby compressing price spreads. The approach adopted in this thesis quantifies this feedback through an iterative price-adjustment framework, applied to the representative weekly price profiles introduced in Section 3.1.1.

3.1.1 Price-representative weeks

The representative-week selection is based on data from 2023-2024 and is performed at weekly resolution to reduce dimensionality while preserving relevant temporal structure. The dataset is first segmented into consecutive

weeks of 168 hours, and each week is assigned a season (Winter/Spring/Summer/Autumn).

Because weeks are compared using a distance metric in the final step, the input features must be in comparable scales/units. The methodology therefore standardizes the underlying variables using z-score normalization.

$$z = \frac{v-\mu}{\sigma} \quad (1)$$

Where v is the original value, and μ and σ are the mean and standard deviation computed over the full 2023-2024 sample. This prevents variables with larger variance from dominating the distance calculation, following common practice in representative-period selection.

After processing, each week w is described by a weekly feature vector $\boldsymbol{x}_w$:

$$\boldsymbol{x}_w = (zPm,w, zPspread,w, zAvgAbsDP,w, zNeg,w, zRLmean,w, zRLspread,w) \quad (2)$$

For each season S, a seasonal centroid vector $\bar{\mathbf{x}}_s$ is computed as the average of the weekly vectors belonging to that season:

$$\bar{\boldsymbol{x}}_S = \frac{1}{|W_S|}\sum_{w\in W_S} \boldsymbol{x}_w \quad (3)$$

Finally, each week's representativeness is measured by its Euclidean distance to the centroid of its own season:

$$w = \|\boldsymbol{x}_w - \bar{\boldsymbol{x}}_s\|_2 = \sqrt{\sum_{k=1}^{6}\left(\boldsymbol{x}_{w,k} - \bar{\boldsymbol{x}}_{s,k}\right)^2} \quad (4)$$

And withing each season, the **representative week** is selected as the observed week with the lowest score. This centroid-distance selection is consistent with distance-based representative-period methods used in energy system modelling (Sisternes & Webster, 2012 and Teichgraeber & Brandt, 2019)

3.1.2 Iterative price-adjustment framework for day-ahead arbitrage

The quantification of storage-induce price changes is performed through an iterative optimisation process applied to each representative week. For a selected representative week, the analysis begins by constructing hourly price vectors consistent with the German day-ahead market design and regulatory framework.

It is worth noting that this analysis is conducted ex-post using historical day-ahead price data, implying full ex-post visibility of hourly prices over the entire representative week. The optimization framework therefore assumes perfect foresight within the weekly horizon. This assumption is standard in retrospective storage studies (Mercier et al. (2023), Lamp & Samano (2022)) and is used to isolate the structural effects of battery arbitrage on price formation from forecasting uncertainty, providing a consistent benchmark for evaluating storage-induced impacts under observed market conditions.

An hourly spot price vector is first determined using observed day-ahead prices for Germany. Based on this vector, two operational price vectors are constructed. The charging price vector represents the effective cost faced by the battery when charging and is defined as the sport price plus applicable network charges and regulated levies. The specific charges applied depend on the regulatory scenario considered, as defined in Section 3.1.4. In contrast, the discharging price vector is defined as the spot price alone. In the German day-ahead market, network charges and regulated levies are applied to electricity consumption, whereas electricity injected into the grid is remunerated at the market price without additional charges. Accordingly, all charges are applied exclusively to the charging leg of battery operation.

The charging and discharging price vectors are introduced into VaasaETT proprietary battery arbitrage optimisation model (SATI), which identifies the profit maximising charging and discharging schedule subject to technical constraints (described in Section 3.1.3). The optimization output specified, for each hour, whether the battery charges or discharges and the corresponding energy volumes. In parallel, indicators such as the number of charging and discharging event and total arbitrage profit are recorded.

To model the impact of battery operation on process, the spot price vector is subsequently updated based on the optimization results. A new spot price trajectory is constructed for the next iteration by modifying hourly prices conditional on battery behaviour. If the battery charges during a given hour, the spot price for that hour is increased to reflect additional demand. If the battery discharges, the spot price is decreased to reflect additional supply. Hours without battery activity remain unchanged.

Price updates follow a band-based adjustment rule designed to ensure meaningful price changes in low-price and negative-price regimes. Percentage-based adjustments become increasingly small as prices approach zero, leading to negligible changes and preventing prices from moving out of near-zero regions. To avoid this effect, a symmetric price band of [-10,+10] EUR/MWh is defined. When prices lie outside the band, adjustments are applied proportionately using a fixed percentage of 10%. When prices lie within the band,

adjustments are applied as a fixed absolute change of ±10 EUR/MWh, ensuring that prices are pushed out of the near zone region before proportional adjustments resume.

Systematically, the spot price update for hour h at iteration n+1 is given by:

$$P_h^{(n+1)} = \begin{cases} P_h^{(n)}(1+0.10), & if\ charging\ and\ P_h^{(n)} > 10, \\ P_h^{(n)} + 10, & if\ charging\ and - 10 \le P_h^{(n)} \le 10 \\ P_h^{(n)}(1-0.10), & if\ charging\ and\ P_h^{(n)} < -10 \\ P_h^{(n)} - 10, & if\ charging\ and\ -10 \le P_h^{(n)} \le 10 \\ P_h^{(n)}, & if\ no\ battery\ activity \end{cases}$$

The updated spot price vector is the used to reconstruct the charging and discharging price vectors, and the optimization is repeated. This iterative process continues until either no charging opportunities remain, or a maximum of 20 iterations is reached. The iteration cap serves as a standardization criterion to endure comparability across scenarios and to prevent excessive iteration depth. Given the assumed initial state of charge of 0.5, at least one arbitrage action may remain feasible even under low-profit conditions; the iteration limit therefore provide a consistent and transparent stopping rule

The resulting price trajectories represent market outcomes in which day-ahead arbitrage opportunities are progressively reduced by the presence of flexible storage, allowing the magnitude and dynamics of storage-induced price changes to be quantified in a controlled and replicable manner.

3.1.3 Model assumptions and input parameters for battery arbitrage analysis

This subsection documents the assumptions and input parameters used in the battery arbitrage optimisation. The objective is to ensure transparency and replicability by clearly stating the modelling inputs and their purpose, while avoiding a detailed description of the internal solution mechanics of the optimization algorithm. The battery is represented as a generic, price-taking asset operating in the German day-ahead market under perfect price foresight, consistent with standard retrospective storage valuation approaches (Lamp & Samano, 2022; Mercier et al., 2023)

Optimisation horizon and temporal resolution

The optimisation is performed over a weekly horizon of 168 hourly time steps, corresponding to one representative week per run. Each representative

week is analysed independently and sequentially under different regulatory charging-cost scenarios. Historical day-ahead prices are used as inputs, implying full ex-post visibility of prices over the optimisation horizon. This set up is commonly adopted in retrospective arbitrage studies to isolate structural price effects from forecasting uncertainty.

All model inputs are expressed at hourly resolutions. Where source datasets were originally available at a 15-minute granularity (e.g. electricity generation by source), hourly values were constructed by averaging the four quarter-hour observations withing each hour. This ensures consistency between price signals and all explanatory variables used in the representative-week framework.

Battery size and duration

A single generic battery configuration is used throughout the analysis to maintain comparability across week and scenarios. The battery is characterised by:

- Power capacity: 2000 MW
- Energy capacity: 8000 MW
- Duration: Fixed at 4 hours, defined as the ratio of energy to power capacity

A fixed battery duration throughout the analysis allows the focus to remain on the interaction between battery operation, charging costs and electricity prices, while avoiding additional complexity from varying technical configurations. Although the modelling framework can accommodate different battery sizes, as single representative configuration is adopted for clarity.

Efficiency and losses

Charging and discharging losses are represented though one-way efficiencies:

- Charging efficiency: 0.85
- Discharging efficiency: 0.85

This implies an effective roundtrip efficiency of approximately **72%**. Modelling efficiencies explicitly on the charging and discharging legs follow standard practice in storage arbitrage models and prevents unrealistically costless intertemporal energy shifting (Lamp & Samano, 2022; Mercier et al., 2023). No self-discharge is assumed, which is appropriate for the weekly optimization horizon considered (Zerrahn et al., 2018; Teichgraeber and Brandt, 2019).

State of charge constraints and boundary conditions

The state of charge (SoC) of the battery is constrained withing the following bounds:

- Minimum SoC: 0.1
- Maximum SoC: 0.9

The initial SoC is fixed at 0.5 at the start of each optimisation iteration withing a representative week. This ensures that all iterations begin from a neutral and identical state of charge, preventing path dependence across iterations and allowing price adjustments to be evaluated independently of prior battery dispatch. The terminal SoC is left unconstrained. As a result, under profit maximisation, the battery typically ends the weekly horizon close to the minimum SoC bound, while always respecting the imposed operation limits. This treatment allows the optimiser to fully exploit within-week arbitrage opportunities without imposing artificial end conditions.

Operational constraints

Simultaneous charging and discharging withing the same hour is not allowed and. This restriction reflects physical feasibility and is a standard feature of arbitrage optimisation formulation (Metz and Saraiva, 2018). The constraint is enforced internally by the SATI optimisation engine and is therefore documented here as a modelling assumption rather than an explicitly coded formulation.

No ramp-rate limits, minimum operating power levels, availability constraints, grid connection limits, or curtailment constraints are included. These simplifications are consistent with a price-taking representation of storage whose purpose is to isolate the effect of price signals and regulatory charging costs, rather than to replicate detailed plant-level or network-constrained dispatch.

Charging and discharging prices

Charging and discharging prices are constructed in accordance with the German regulatory framework. Network charges and regulated levies apply to electricity consumption and are therefore included exclusively in the charging price, while discharging revenues are based solely on the day-ahead spot price.

Variable operating cost and profit calculation

A variable operating cost of €0.25/MWh is specified as an input parameter in the SATI environment to represent marginal cycling-related costs. However, the model outputs do not provide an explicit decomposition of the optimisation objective, nor report net revenues or cost-adjusted profits. Consequently, arbitrage revenues are calculated ex-post using charging and discharging quantities and their corresponding tariffs, without explicitly subtracting the operating cost term. Reported profits therefore represent gross arbitrage revenues, which is appropriate given that the focus of the analysis is on price dynamics and arbitrage saturation rather than on private profitability.

Price caps and market constraints

Market price caps are not imposed as explicit constraints in the optimisation model. They are reflected only insofar as they are present in the historical day-ahead price data. This ensures that the optimisation is evaluated under observed market conditions without introducing non-observed constraints.

3.2 Data collection and processing

This section describes the data sources used in the analysis and the preprocessing steps applied prior to their use in the methodology presented in Section 3.1. The study relies exclusively on historical data for Germany covering the year 2023-2024, and all datasets are harmonised to an hourly resolution to ensure consistency across price, load, and generation variables.

3.2.1 Data source

Electricity prices

Hourly German day-ahead electricity prices (EUR/MWh) were obtained from the Energy Market Price platform, with access provided by VaasaETT. The platform aggregates market-clearing prices from European power exchanges. The dataset covers Germany only (DE) and corresponds to realised market outcomes in the day-ahead market for the year 2023-2024.

Electricity demand and generation

Hourly total electricity load and generation (by source) data were retrieved from ENTSO-E Transparency platform. Data were collected at country level for Germany, rather than at bidding-zone level, to exclude Luxemburg and maintain consistency with the geographical scope of the price data. Generation data include all reported technologies later used to construct the residual load indicator.

Fuel and emission prices

Fuel and emission cost data were also obtained from Energy Market Price platform. This includes prices of natural gas, coal, and EU Emission Trading System (EUA – European Union Allowance), expressed in their respective market units. These data are used as explanatory variables and contextual input in the broader analysis.

All data sets are provided in UTC time as supplied by the original sources. No explicit adjustments were made for daylight saving time.

3.2.2 Data processing

To ensure consistency across datasets, all-time series were converted to a common hourly resolution. Where original data were available at 15-minute granularity (notably generation data), hourly values were constructed by calculating the arithmetic mean of the four quarter-hour observations withing each hour. The resulting hourly value was assigned to the corresponding hour timestamp.
Missing hourly observations were handled though interpolation, ensuring continuous time series for all variables used in the analysis. After pre-processing, all datasets consist of fully aligned hourly observations for the period 2023-2024.

3.2.3 Constructed Variables

Several variables used in the analysis are not directly observed but are derived from the processed hourly data. These variables are constructed at weekly level and serve as inputs to the representative-week selection described in Section 3.1.1. Definitions and units are summarized below:

- Weekly average electricity price (EUR/MWh) – The arithmetic mean of hourly day-ahead electricity prices withing a calendar week.
- Weekly price spread (EUR/MWh) – The difference between the maximum and minimum hourly electricity prices observed withing a week.
- Share of negative prices (%) – Proportion of hourly observations within a week for which the electricity price is below zero.
- Average absolute price change (EUR/MWh) – Weekly average of the absolute hour-to-hour price difference, defined as

$$\frac{1}{N-1}\sum_{t=2}^{N}|P_t - P_{t-1}| \quad (5)$$

Where P_t denotes the hourly electricity price

- Residual load mean (MWh) – Weekly average of residual load, defined as total electricity load minus renewable electricity generation.
- Residual load spread (MWh) – Difference between the maximum and minimum residual load values observed within a week.

All constructed variables are derived directly from the hourly datasets described in Sections 3.2.1 and 3.2.2. The exact formulation and calculations are documented in the accompanying Excel files.

Peak/Valley hour sets and level-effect metrics

To quantify price level effects in high- and low-price conditions, "Peak hours" are defined as the set of hours for which the Spot price is greater than or equal to the 90th percentile of Spot prices within the presentative week $\{h: P_{0,h} \geq P90(P_0)\}$. "Valley hours" are defined analogously as the set of hours in which the Spot price is less than or equal to the 10th percentile $\{h: P_{0,h} \leq P10(P_0)\}$. Percentiles are computed using the Spot price distribution of each representative week, and the resulting peak/valley hour sets are held fixed when evaluating converged prices under each charging-cost structure.

Peak shaving is measured as the change in the average price withing Spot-defined peak hours between Spot and the converged price vector:

$$\Delta P^{peak} = P'_{\{h:P_{0,h} \geq P90(P_0)\}} - P_{0\,\{h:P_{0,h} \geq P90(P_0)\}} \quad (6)$$

Valley filing is measured similarly withing Spot-defined valley hours:

$$\Delta P^{valley} = P'_{\{h:P_{0,h} \leq P10(P_0)\}} - P_{0\,\{h:P_{0,h} \leq P10(P_0)\}} \quad (7)$$

By construction, negative values of ΔP^{peak} indicate a reduction in high-price hours (peak shaving), while positive values of ΔP^{valley} indicate an increase in low-price hours (valley filling).

3.2.4 Data scope and limitations

The analysis is based on historical data aggregated to hourly resolutions and country-level spatial scope. Using country-level load and generation data excludes intra-zonal congestion effects and cross-border interactions, which are therefore not explicitly represented. In addition, the use of interpolated values for missing observations may smooth short-term variability. These limitations are considered acceptable given the focus of the study on price

dynamics, representative periods, and system-level effects of battery energy storage.

3.3 Battery operation under different bill structures

This section defines the charging-cost cases applied to BESS in the modelling framework. The cases are derived directly from the battery charging components and exemptions discussed in Section 2.7 and represent alternative billing treatments under which batteries are assumed to consume electricity from the grid.

All cases are evaluated using the same historical data, battery parameters, and optimisation framework described in Section 3.1 and 3.2. The only element that varies across cases is the set of charges applied to electricity consumed during battery charging. No charges are applied to electricity injection to the grid across all cases.

3.3.1 Definition of charging-cost cases

Three charging-cost cases are considered:

- **Case 1: Full exemption** – Batteries are exempt from network charges and regulated levies when charging, with only concession-related fees applied. This case represents the lowes possible charging cost consistent with the German billing framework.
- **Case 2: Partial exemption** – Batteries remain subject to network charges but are exempt from regulated components. Charging costs are therefore higher than in the full exemption case, but lower than under full cost exposure.
- **Case 3: No exemption** – Batteries face the full set of applicable network charges and regulated levies when charging. This case represents the most restrictive charging-cost treatment.

These cases correspond to the range of charging-cost treatment identified in Section 2.7 and are applied consistently across all representative weeks and iterations.

3.3.2 Purpose of charging-cost comparison

The comparison between charging-cost cases allows the analysis to isolate how regulatory billing conditions influence battery arbitrage behaviour and the resulting compression of day-ahead price spreads. By holding all other assumptions constant, differences in price outcomes across cases can be attributed directly to the charging-cost treatment applied to batteries.

4 Results

This section presents the results of the iterative price-adjustment analysis applied to representative weeks of the Germany day-ahead electricity market. These results quantify how BESS arbitrage modifies day-ahead price formation under alternative charging-cost structures. Results are presented descriptively, without interpretation of policy or investment implications, and are discussed in Section 5.

4.1 Representative weeks and baseline characteristics

The baseline characteristics of the selected representative weeks that characterize the 2023-2024 German day-ahead (DA) market and used in this analysis are summarized in Table 2.

Table 2. Descriptive statistics of representative weeks

	Week 49 - Winter	Week 21 - Spring	Week 29 - Summer	Week 37 - Autumn
Mean price (EUR/MWh)	85.55	85.54	72.98	79.51
Max price (EUR/MWh)	146.3	221.5	227.6	300
Min price (EUR/MWh)	26.4	-23.9	-11.5	-0.5
Price spread (EUR/MWh)	119.9	245.3	239.1	300.5
Negative hours (%)	0.0	3.6	6.5	5.4
Max Δ Price (EUR/MWh)	25.2	61.8	97.4	154.8
Average Residual Load (MW)	39966	27420	27345	28838
Residual Load spread (MW)	53277	48117	50931	49132
Residual load (%)	64	58	55	56

The selected sample span seasons and markedly different price regimes, ranging from a relatively stable winter week (W49) to weeks characterized by sharp price spikes and negative-price episodes (W21, W29 and W37). Baseline weekly spreads vary widely from 120 EUR/MWh in W49 to 300.5 EUR/MWh in W37, indicating substantial variability in arbitrage potential across the sample.

To visualise the different price structures captured by the selected weeks, Figure 2 plots the hourly Spot price profiles (P_0) for Weeks 49, 21, 29 and 37.

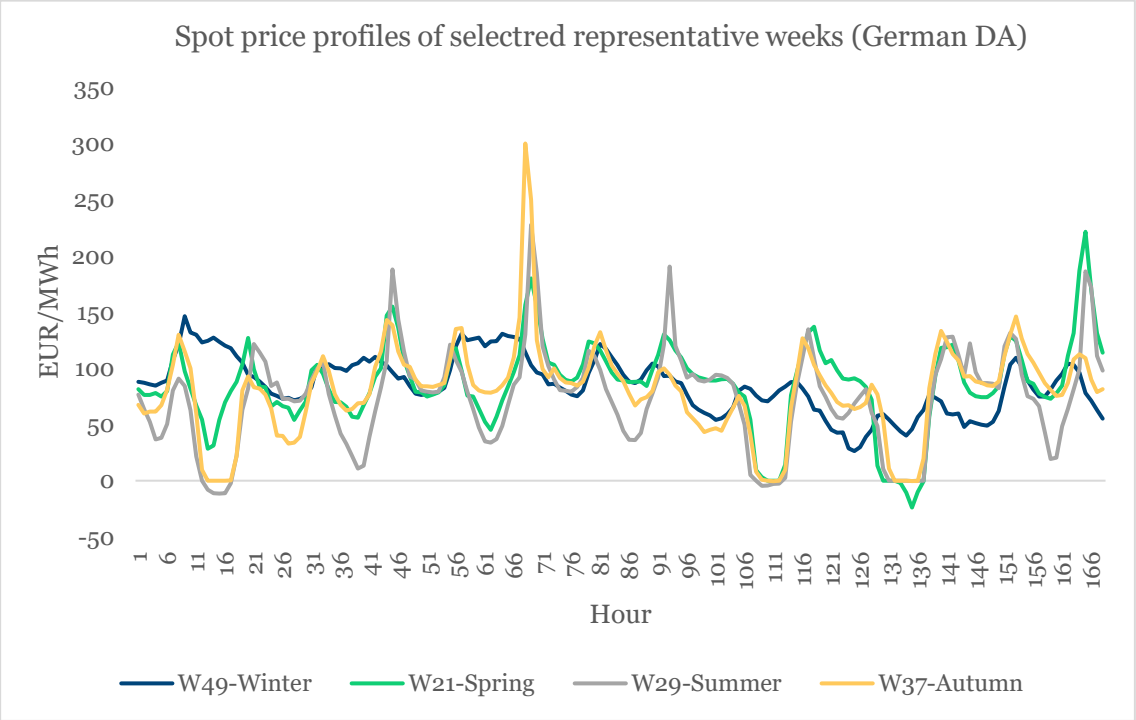


Figure 2. Spot price profiles of selected representative weeks (Germany Day-Ahead 2023-2024, EUR/MWh) (Source: Energy Market Price, 2025)

Residual load levels and spreads differ across weeks, supporting the interpretations of BESS impacts as conditional on underlying system tightness and renewable-driven volatility.

4.2 Battery behaviour under different charging-cost structures

Dispatch results are reported at the final economic dispatch, defined as the last iteration in which the battery performs a complete arbitrage cycle with both non-zero charging and non-zero discharging. This avoid reporting artefacts caused by the initial state of charge (50%), which can allow residual discharge even when no profitable charging remains. Results are shown for

three charging-cost structures (Full exemption, No exemption, Network exemption) and across all four representative weeks.

4.2.1 Temporal distribution of battery charging and discharging at final economic dispatch

Charing and discharging timing are grouped into three time-of-day periods: Night (22:00-05:59), Midday (10:00 15:59) and Other (all remaining hours). The time-of-day allocation of charging differs strongly by week and charging -cost structure. In W49, charging is concentrated entirely during Night hours (100%) across all regimes, as shown in Figure 3), consistent with this week's relatively "classic" overnight low-price pattern. In contrast, in the spring and autumn weeks, charging shifts towards Midday:

- In W21, charging is 73% Midday and 27% Other under Full exemption and 100% under No exemption.
- In W37, charging is 100% Midday under Full and No exemption, while Network exemption yields a more mixed pattern (47% Midday/53% Other).
- A key exemption is W29 under Network exemption, where charging remains 100% during the Night period, indicating that under this cost regime the "cheapest" charging opportunity revert to night hours for this specific price shape.

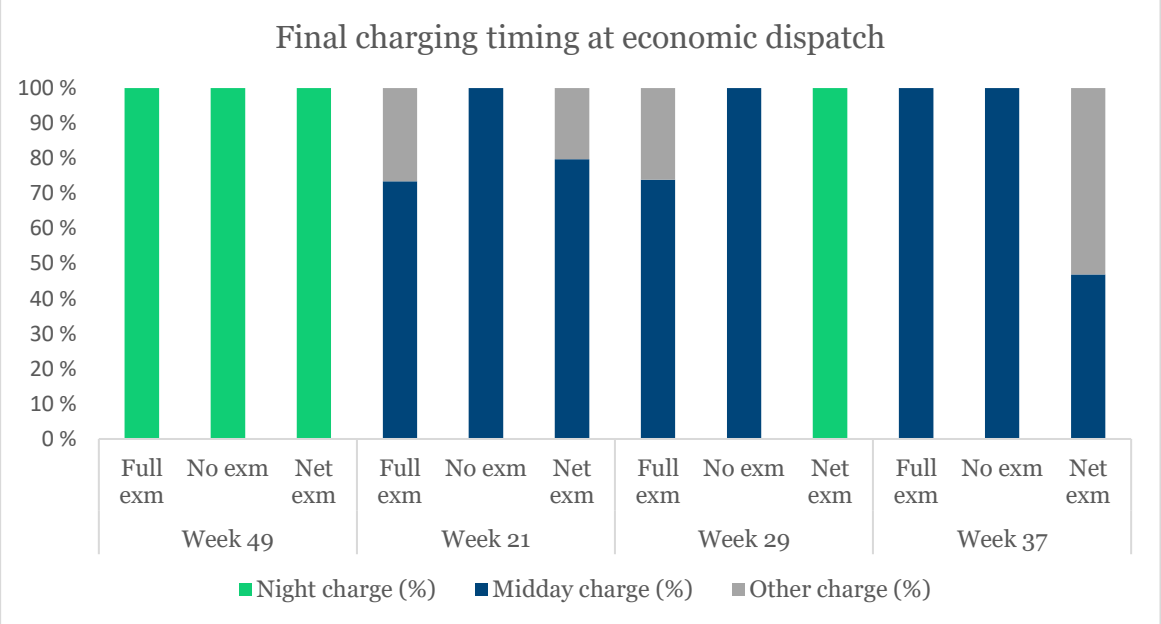


Figure 3. Final charging time-of-day distribution at final economic dispatch (energy shares; Night 22:00-05:59, Midday 10:00-15:59, Other remaining hours)

Discharging is dominated by the Other hours period in most scenarios indicating that the highest-price hours frequently occur outside the Midday and Night windows (typically morning, evening ramps and spikes). Discharging patterns are more heterogenous than charging, as illustrated by Figure 4. For example:

- In W49 Full exemption, discharging splits between Night (65%) and Other (35%), while No exemption shifts discharging entirely to Other.
- In W29 Full exemption, discharging spreads across Night (40%), Midday (16%) and Other (44%), consistent with the presence of multiple peak periods withing the week.

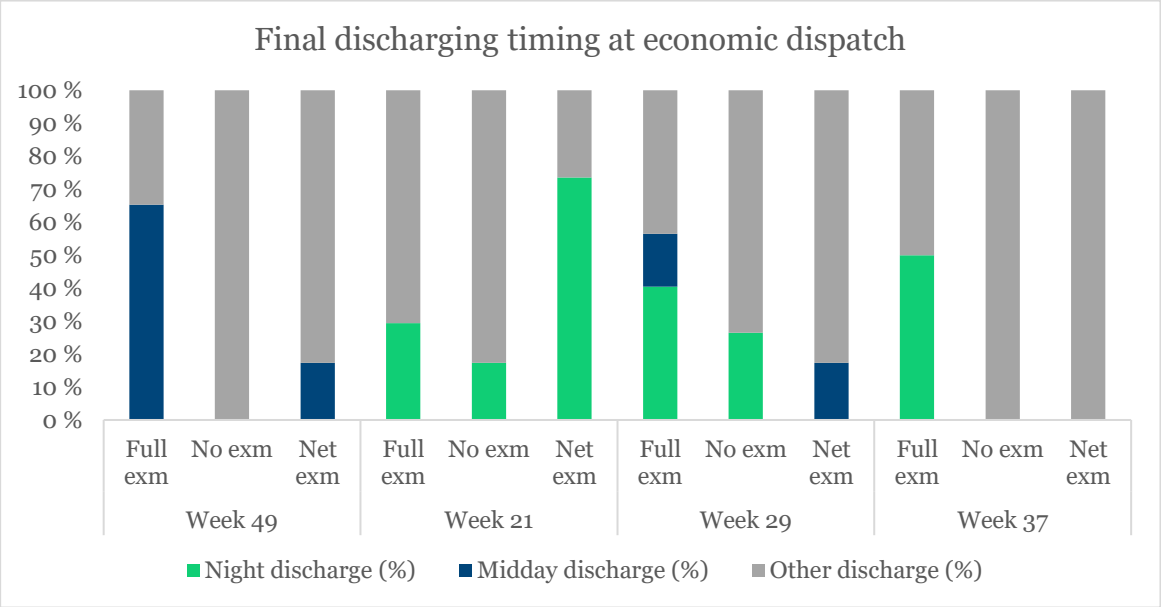


Figure 4. Final discharging time-of-day distribution at final economic dispatch (energy shares; Night 22:00-05:59, Midday 10:00-15:59, Other remaining hours)

Overall, the figures show that charging-cost structures influence not only how much the battery moves, but also when it provides flexibility, especially on the charging side.

4.2.2 Dispatch intensity and activity at final economic dispatch

A comparison between charged and discharged energy at the final economic dispatch across all scenarios is provided in Figure 5.

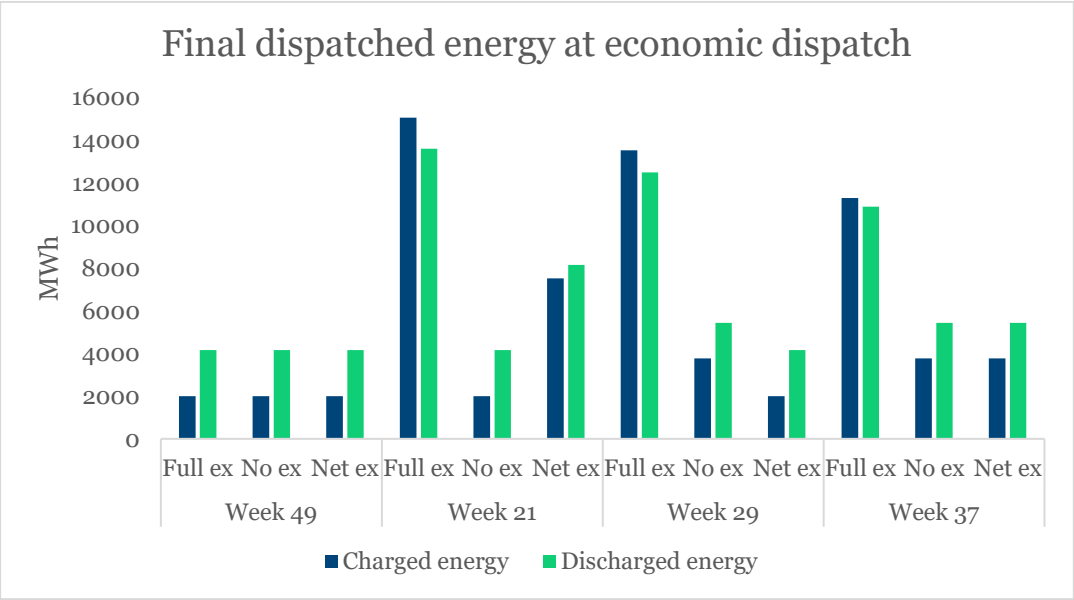


Figure 5. Final dispatched energy at final economic dispatch by week and charging-cost structure (charged vs discharged energy, MWh)

Under full exemption, the battery is highly active in volatile weeks, charging just over 15 GWh in W21, slightly more than 13.5 GWh in W29 and almost 11.3 GWh in W37, compared to only 2 GWh in W49. NO exemption systematically limits dispatched energy volumes with charged energy in W49 and W21 at 2 GWh and 3.7 GWh during in W29 and W37. Network exemption produces intermediate outcomes: 7.7 GWh charged in W21, 2 GWh in W29 and 3.7 GWh in W37.

These differences in intensity are mirrored in active hours. Under Full exemption, charging expands to 8 hours (W21), 7 hours W29 and 6 hours (W37), compared to 1 hour in W49, as shown in Figure 6. In contrast, under No exemption charging is constrained to 1-2 hours depending on week, consistent with the lower energy volumes dispatched seen above. This shows that lower charging penalties do not only increase volume; they also broaden participation across more hours, which helps explain why they deliver larger price impacts in Section 4.3 and 4.4.

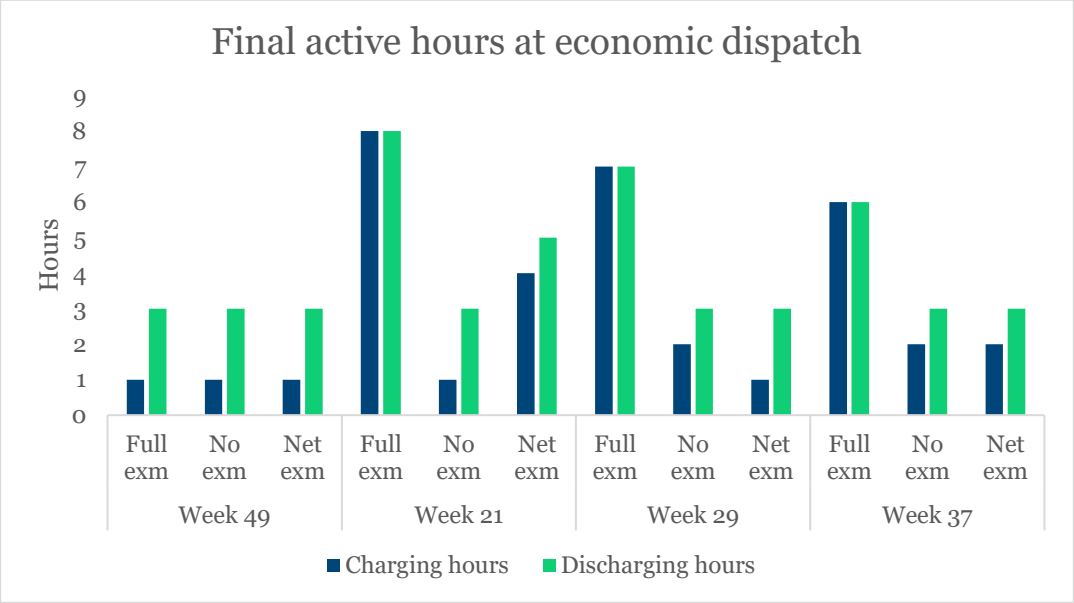


Figure 6. Final active charging and discharging hours at final economic dispatch by week and charging-cost structure (hours).

4.3 Impact of BESS arbitrage on day-ahead price levels

As defined in Section 3.2.3, "peak hours" are identified as $Spot \geq P90\ (Spot)$ and "valley hours" as $Spot \leq P10\ (Spot)$ within each representative week. Impacts are evaluated by comparing converged prices (P') to Spot (P_0) over those same Spot-defined hours, ensuring comparability across charging-cost structures.

4.3.1 Definition of high- and low-price hours

To quantify level effects, "high-price hours" are defined as hours in which Spot ≥ P90 (Spot) for the representative week, and "low-price hours" as hours in which Spot ≤ P10 (Spot). Peak shaving is measured as the difference between the average converged price in P90-hours and the average Spot price in those same hours. Valley filling is measured analogously for P10-hours. Using Spot-based thresholds ensured the evaluation set is consistent across cost structures.

Figure 7 illustrates the flattening mechanism: compared to Spot, the converged curves reduce sharp peaks and lift low-rice periods, with the magnitude of smoothing different across charging-cost structures.

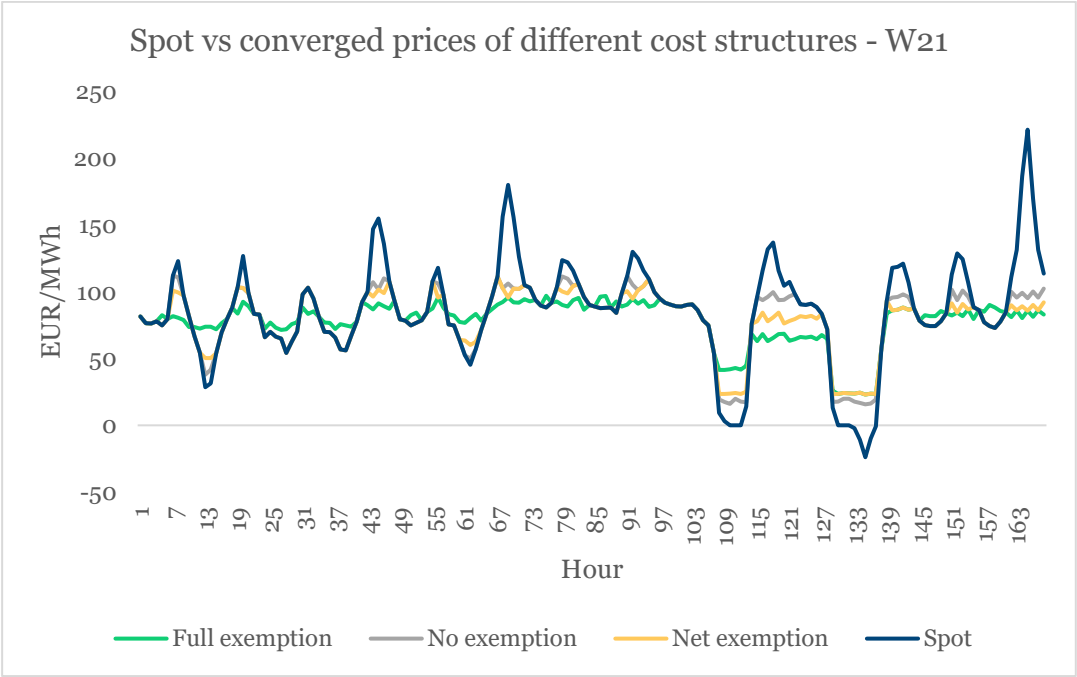


Figure 7. Week 21 illustrative comparison: Spot (P_0) vs converged prices (P') under Full, Network and No exemption (EUR/MWh).

4.3.2 Peak shaving and valley filling results

The magnitudes of the smoothing effect vary substantially by week and charging-cost structure, as seen from Figure 8. Peak shaving results are presented as negative prices, due to their reductive effect, while valley filling are shown as positive, indicating a price increase.

Week 49 shows relatively small impacts, reducing peak prices as little as -4.7 EUR/MWh in the No exemption structure, and producing the highest impact in the Full exemption scenario, reducing the price by -25 EUR/MWh. Valley filling ranged between +0.2 and +12.5 EUR/MWh in the No and Full exemption scenario, respectively.

In the more volatile weeks, impacts rise sharply. In W21, peak shaving reaches -64.2 EUR/MWh (Full), compared to -49.1 and 57.2 EUR/MWh in No and Network exemption, respectively. Valley filling is +33.3 EUR/MWh (Full) versus +17.4 and 23.9 EUR/MWh in No and Network exemption scenarios, correspondingly.

Week29 exhibits the strongest peak shaving under Full exemption (-75.5 EUR/MWh), with -54.3 and -68.4 EUR/MWh in No and Network exemption, respectively, while valley filling is +33.9, +19.8 and +25.1 EUR/MWh in Full, No and Network exemption structures, respectively.

In W37, the highest impact on peak shaving and valley filling is under Full exemption, with a price increase of +41.1EUR/MWh and a peak shaving of -65.2 EUR/MWh.

Across all weeks, Full exemption delivers the largest absolute peak shaving and valley filling, while No exemption yields the smallest. Network exemption typically sits between the two, indicating that partial relief of charging costs preserves a substantial portion of the price-level impact.

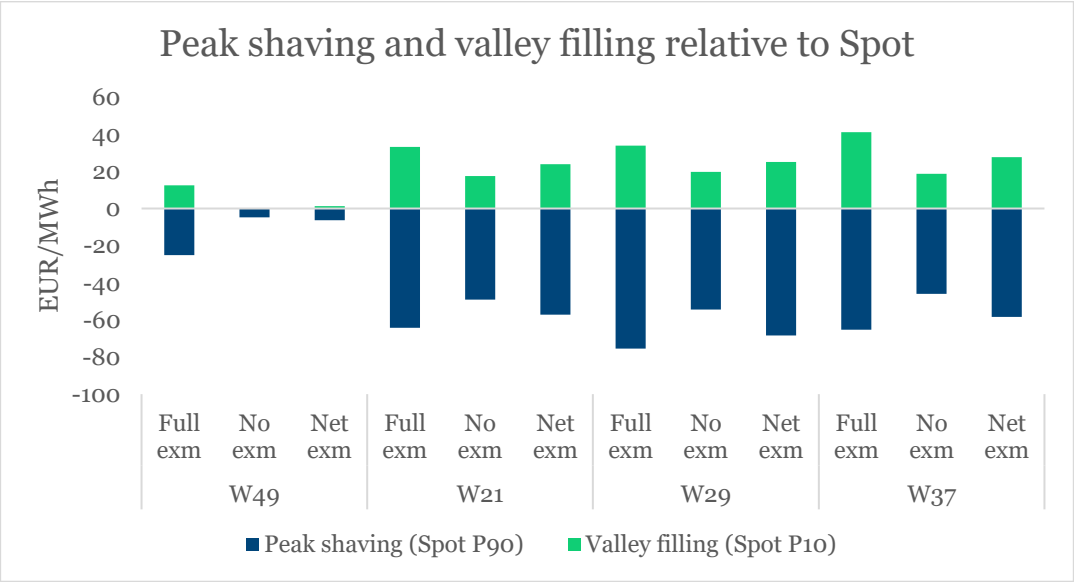


Figure 8. Tail-hour price effects relative to Spot: peak shaving in Spot P90 hours and valley filling in Spot P10 hours (EUR/MWh).

4.4 Structural price changes

While section 4.3 isolated tail-hour level effects (P90/P10), this section examines whole-distribution outcomes that reflect the saturation/flattening of arbitrage opportunities: Weekly spread compression and volatility reduction relative to Spot.

4.4.1 Weekly spread compression

Weekly spreads decline sharply from Spot to converged prices across all charging-cost structures (Figure 9). The largest absolute compression occurs in the most volatile weeks (Week 21, 29 and 37). Full exemption produces the strongest spread reduction, while No exemption reduces spreads less, reflecting lower charging volumes and weaker ability to absorb low-price hours and supply high-price hours.

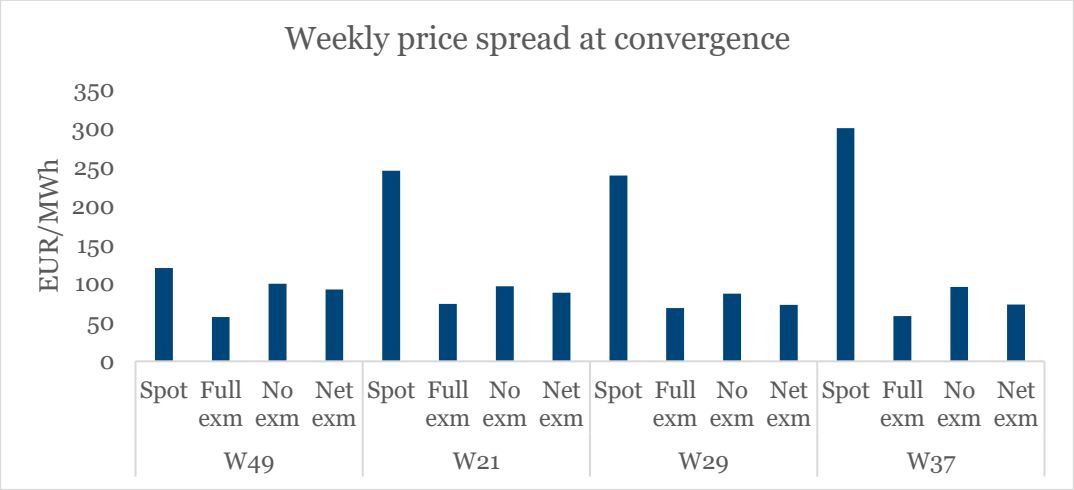


Figure 9. Weekly price spread in Spot and converged prices by week and charging-cost structure (EUR/MWh).

4.4.2 Volatility reduction

Volatility is measured as the reduction in mean absolute hourly price change relative to spot, falls substantially under all regimes, with Full exemption producing the largest volatility reductions in each representative week, as illustrated by Figure 10. Results are presented as percent change relative to Spot.

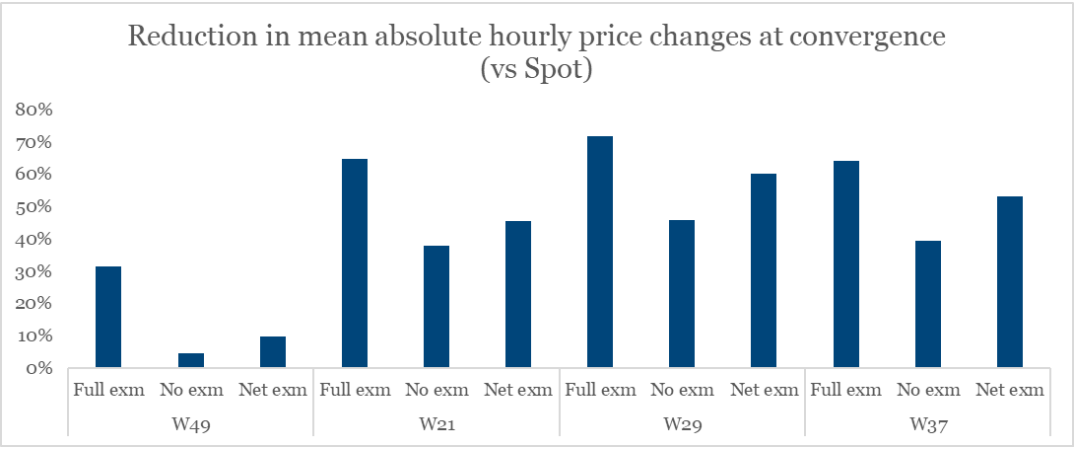


Figure 10. Volatility reduction relative to Spot: change in mean absolute hourly price difference at convergence (%, by week and charging-cost structure).

The pattern mirrors the spread results: structures enabling higher dispatch intensity (Section 4.2) also deliver stronger compression of short-term price variability.

4.5 Comparison of price impacts across charging-cost structures

Across all results dimensions, behavioural (dispatch intensity and timing) and price-based (P90/P10, spreads, volatility), charging-cost structures systematically rank the magnitude of impacts. Full exemption consistently produces the strongest effects, including the largest peak shaving (up to -75.5 EUR/MWh in W29) and valley filling (up to +41.1 EUR/MWh in W37), as well as the strongest structural compression (e.g., spreads reduced to 58,3 EUR/MWH in W37, and volatility reduced by 72% in W29) (Figures 7-9). No exemption yields the weakest impacts, while Network exemption typically lies between Full and No, with some week-specific deviations driven by differences in where the cheapest charging hours occur (e.g., night vs midday).

4.6 Summary of key results.

Across all representative weeks, converged prices exhibit a clear flattening effect: high-price hours are reduced (peak shaving), low-price hours are raised (valley filling), and the overall distribution becomes more compressed (low weekly spreads and lower volatility). The magnitude of these results depends strongly on both the week-specific baseline regime and the charging-cost structure, with Full exemption consistently enabling the largest dispatch intensity and the strongest price impacts.

5 Summary/Conclusions

5.1 Research objective and approach

This thesis analysed how large-scale battery energy storage systems (BESS), operating though day-ahead (DA) electricity arbitrage, can modify German DA price formation, and how the magnitude of these price effects depends on regulatory charging-cost structures. The focus was explicitly placed on price dynamics and market outcomes, rather than on storage profitability or revenue optimization.

To isolate storage-induced price effects, the thesis developed a price-based analytical framework that combines a profit-maximizing, price-taking arbitrage dispatch (VaasaETT SATI) with an iterative price-adjustment mechanism. Starting from historical German DA Spot prices, the framwork repeatedly (i) determines optimal charging and discharging behaviour under technical constraints and a given charging-cost structure, and (ii) updates hourly prices conditional on battery operation, reflecting additional demand during charging and additional supply during discharging. This process continues until arbitrage opportunities are exhausted or a maximum iteration cap is reached.

The analysis is conducted **ex post**, assuming perfect price foresight within a weekly horizon, and results are reported at the **final economic dispatch**, defined as the last iteration with both non-zero charging and non-zero discharging. This reporting choice avoids a discharge-only terminal artefact arising from the fixed initial state of charge. Empirical results are derived from four representative German DA weeks in 2023-2024, evaluated under three charging-cost structures: Full exemption, Network exemption (network charges only), and No exemption. All timestamps are expressed in UTC.

5.2 Main findings

5.2.1 Arbitrage behaviour depends on week conditions and charging-cost structure

The dispatch results demonstrate that BESS arbitrage behaviour is highly dependent on both the underlying price regime of the week and the regulatory treatment of charging. Week characterised by deeper price valleys, higher peaks and larger intertemporal spreads enable more intensive arbitrage operation, with higher energy shifted and broader participation across hours.

Charging-cost structures materially shape this behaviour. Lower charging penalties allow the battery to shift more energy and to charge across a wider

range of hours, while higher charging costs restrict both the intensity and temporal scope of arbitrage. These behavioural differences provide the primary mechanism explaining why price impacts differ across regulatory scenarios.

5.2.2 BESS arbitrage reshapes the DA price curve though peak reduction and valley uplift

Across all representative weeks, converged price trajectories show a systematic reshaping of the DA price curve consistent with arbitrage saturation. High-price periods are reduced through discharging activity, while low-price periods are lifted though charging. The strongest tail-hour impacts occur in the weeks with the most pronounced baseline volatility and price dispersion, confirming that underlying market conditions remain a key driver of impact magnitude.

Charging-cost design strongly affects the extent of these price-level effects. Full exemption consistently delivers the largest reduction in high-price hours and the strongest uplift of low-price hours. No exemption produces substantially weaker tail-hour effects, while Network exemption typically yields intermediate outcomes. These results indicate that regulatory charging-cost structures can materially influence how strongly storage modifies DA prices, even when market conditions are otherwise favourable to arbitrage.

5.2.3 Structural price compression: spreads and volatility decline at convergence

Beyond tail-hour effects, the results show that large-scale BESS arbitrage can materially compress the overall weekly price distribution. Converged prices exhibit lower weekly max-min spreads and reduced hour-to-hour price variability relative to Spot price. This structural compression reflects the progressive reduction of intertemporal price differences as arbitrage opportunities are internalised though iterative storage operation.

Across all representative weeks, the ranking of outcomes by charging-cost structures is consistent with the behavioural results: Full exemption yields the strongest compression of spreads and volatility, Network exemption intermediate effects, and No exemption the weakest. This consistency reinforces the interpretation that charging-cost design influences both how storage operates and how strongly it reshapes price dynamics.

5.3 Implications

5.3.1 Implications for market outcomes under increasing storage deployment

The results provide controlled evidence that sufficiently deployed BESS operating through DA arbitrage can act as a **price-smoothing mechanisms** in the German DA market. By reducing extreme prices and compressing intertemporal spreads, storage operation can moderate short-term volatility and reshape hourly price trajectories, effects that are particularly pronounced in volatile, high-renewable markets.

Because the analytical framework explicitly internalises the feedback between storage operation and prices, the results capture a key dynamic of large-scale flexibility deployment: price impacts are inherently **self-limiting**, as arbitrage activity reduces the very price differences that initially motivate it.

5.3.2 Implication for arbitrage value and revenue cannibalization

Although this thesis does not quantify arbitrage revenues, the observed compression of price spreads and reduction in short-term volatility imply a **progressive erosion of arbitrage opportunities** as storage-induced price smoothing strengthens. This mechanism is consistent with the revenue cannibalization effect described in the storage literature, whereby increasing storage participation narrows intertemporal price differentials and reduces the remaining arbitrage value available to storage assets.

In this thesis, cannibalization is not measured directly but is reflected qualitatively in the converged price trajectories: the scenarios that deliver the strongest system-wide price smoothing are also those in which arbitrage opportunities are most fully exhausted. This highlights an inhered trade-off between the market-wide benefits of price smoothing and the persistence of arbitrage value for additional storage capacity.

5.3.3 Regulatory relevance of charging-cost design

A central contribution of this thesis is to demonstrate that the regulatory treatment of charging influences not only battery dispatch behaviour but also the magnitude of storage-induced price effects. Full exemption enables the strongest arbitrage response and the largest price-smoothing outcomes. Network exemptions, limited to the removal of network charges only, typically preserves a significant share of these effects, while No exemption materially weakens them by constraining charging and reducing arbitrage activity.

These findings indicate that charging-cost regulation is not neutral with respect to DA price formation. Regulatory design choices can therefore influence not only storage economics but also the extent to which storage modifies wholesale price dynamics in power systems with high shares of VRES.

5.4 Limitations and uncertainties

This thesis adopts a deliberate controlled framework to isolate storage-induced price effects. Several limitations define the interpretation boundary of the results:

1. **Perfect foresight (ex post) assumption**. The optimisation assumes full knowledge of hourly process withing each week. In practice, forecast uncertainty would reduce dispatch optimality and likely dampen price impacts.
2. **Reduced-form price adjustment**. Hourly price updates are implemented through a stylised band-based rule rather than through full market-clearing with supply curves, unit commitment, or strategic bidding. Results therefore represent internally consistent price feedback rather than equilibrium outcomes.
3. **Parameter sensitivity**. Convergence dynamics may depend on the chosen adjustment parameters and iteration cap. Alternatively parameterisation could yield different magnitudes of final price compression.
4. **Representative-week scope**. Only four weeks from 2023-2024 are analysed, limiting the ability to generalise results across all market conditions.
5. **Generic battery representation**. The analysis uses a single 2000 MW/4-hour battery configuration with simplified operational constraints, which may overstate flexibility relative to real-world assets.
6. **Boundary conditions on state of charge**. Fixing initial SoC at 0.5 and leaving terminal SoC unconstrained enables full exploitation of within-week arbitrage but may overstate end-of-horizon discharging relative to operational strategies that preserve reserve SoC.
7. **Revenue analysis out of scope**. Although the results imply diminishing arbitrage opportunities as prices are smoothed, arbitrage revenues are not quantified. Revenue cannibalization is therefore discussed only qualitatively, based on observed price compression and established literature.

5.5 Recommendations for future work

Future research could extend this analysis by:

- Performing **sensitivity analyses** on the price-update mechanism and stopping criteria;
- Analysing **different battery sizes and durations** to identify saturation thresholds for price impacts;
- Expanding the empirical scope to **full-year datasets** and additional years;
- Integrating simplified **market-clearing representations** to bridge towards equilibrium outcomes;
- Extending the framework to **multi-market operation**, including intraday and balancing services.

5.6 Concluding statement

This thesis shows that large-scale BESS operating through day-ahead arbitrage can materially reshape German DA price formation by reducing high-price extremes, lifting low-price periods, and compressing both price spreads and short-term volatility. The magnitude of these outcomes depends not only on underlying market conditions but also strongly on the regulatory charging-cost structure faced by storage. Full exemption produces the strongest dispatch response and the largest price impacts, Network exemptions typically yield intermediate outcomes, and No exemption consistently weakens arbitrage saturation and price smoothing. These findings demonstrate that charging-cost regulation influence both storage operation and the extent to which storage modifies wholesale price dynamics in high-renewable power systems.

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